

Star Vetter - confirmed eclipsing binary candidates

TIC	Period (d)	Depth	BLS SNR	Pixel SNR	Offset	Tier
437150434	5.2853	15.003%	1081	85.33	1"	Clean
237115819	14.7921	8.883%	507	27.82	4"	Clean
66732476	3.1087	9.814%	370	26.93	8"	Clean
29495157	5.1474	8.579%	333	52.62	5"	Alias
119459681	5.9442	15.977%	311	68.15	2"	Clean
411434067	1.3536	4.289%	193	33.70	4"	Alias
330901906	3.7321	0.498%	60	87.24	4"	Thin margin
303470667	5.4319	2.220%	57	5.02	9"	Marginal
273158289	1.1519	1.255%	41	17.24	2"	Thin margin
111149941	5.2401	0.045%	10	9.73	21"	Ambiguous
251898081	27.3089	18.630%	5535	464.16	3"	Alias
467214801	0.7609	0.950%	2134	21.36	0"	Alias
399770041	1.5802	4.291%	489	36.92	1"	Clean
349701853	30.3427	28.445%	343	51.05	6"	Clean
233459038	25.3384	8.315%	237	84.46	5"	Alias
281310348	1.7702	7.273%	180	57.87	2"	Clean
318887646	14.2817	3.605%	150	6.94	13"	Clean
444465823	18.5133	3.476%	147	25.18	1"	Alias
380943251	0.7209	7.667%	101	15.26	3"	Clean
66716482	8.0283	3.329%	65	21.51	7"	Clean
309656476	28.9204	0.050%	25	13.80	14"	Clean
229494833	15.3839	0.037%	15	5.19	12"	Alias
44533527	14.5952	0.101%	12	7.33	16"	Clean
220423668	3.6511	4.525%	755	17.67	6"	Alias
365998904	1.2494	2.398%	375	15.52	3"	Clean
233452636	1.9063	5.467%	202	8.94	2"	Thin margin

26 of 227 novel candidates confirmed on-target by TESS pixel-level difference imaging.
Generated by f5_dossier.py from f3i_vet.py, f4f_novelty.py and diag_pixel.py results.
Period column is the BLS-fitted value; candidates tiered Alias have a doubled true period - see their page.

TIC 220423668

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 3.651055 d (ALIASSED, see below)
-> corrected period: 7.302111 d
Epoch (t0) 1413.9498 BTJD
Duration 0.1200 d
Depth 4.525 % (45,247 ppm)
SNR 755.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

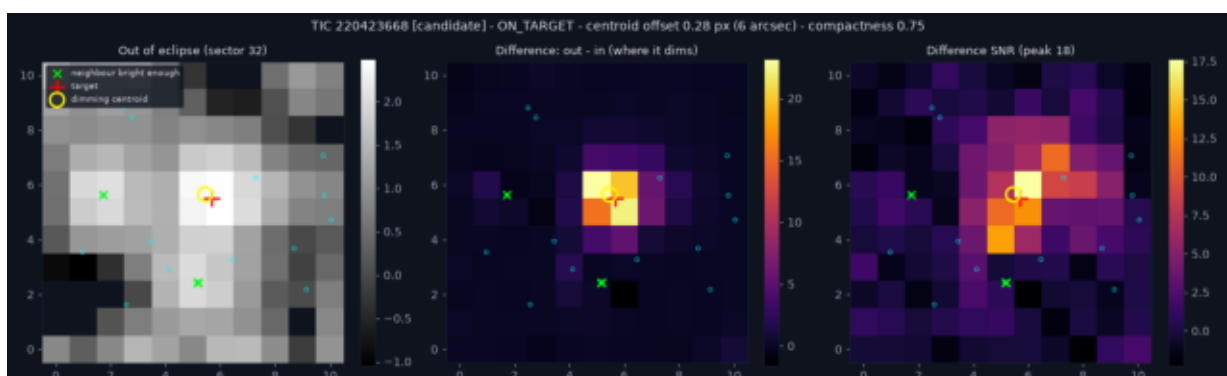
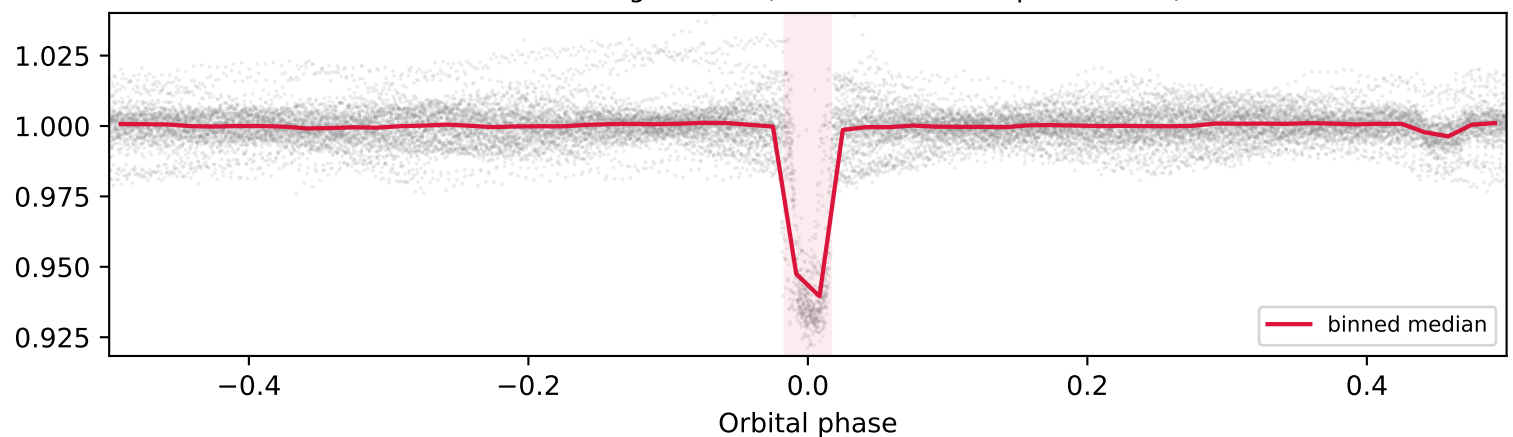
Eclipses observed 55 (gate ≥ 3)
Robust depth ratio 1.27 (gate ≥ 0.50)
Robust depth sigma 215.5 (gate ≥ 7.0)
Odd/even z-score 8.94 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 32 light-curve dip in this sector: 83.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 17.67 (gate ≥ 5.0) Compactness 0.75 (gate ≥ 0.50)
Dimming centroid offset 0.28 px (6 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 220423674
3.23 px away, dTmag +1.58 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 8.94 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 7.302111 d, in any write-up.

TIC 365998904

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.249358 d
Epoch (t0) 1654.4709 BTJD
Duration 0.1200 d
Depth 2.398 % (23,984 ppm)
SNR 375.0

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

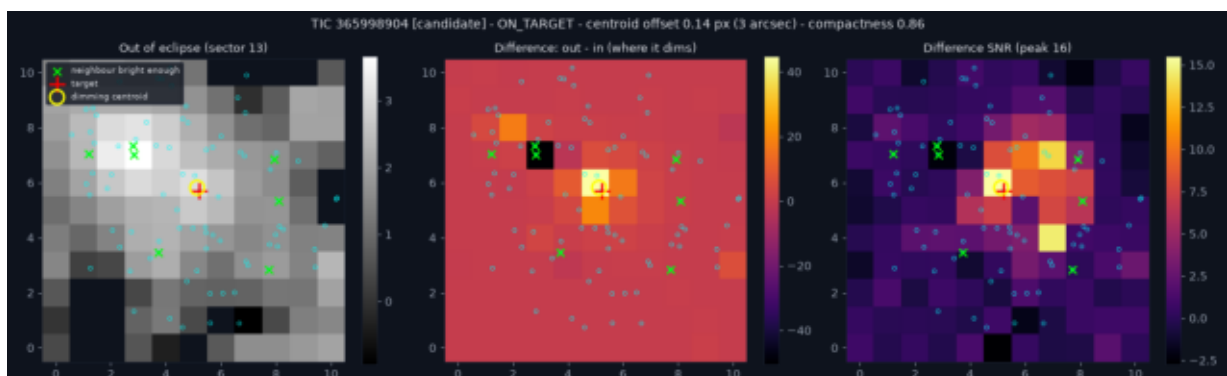
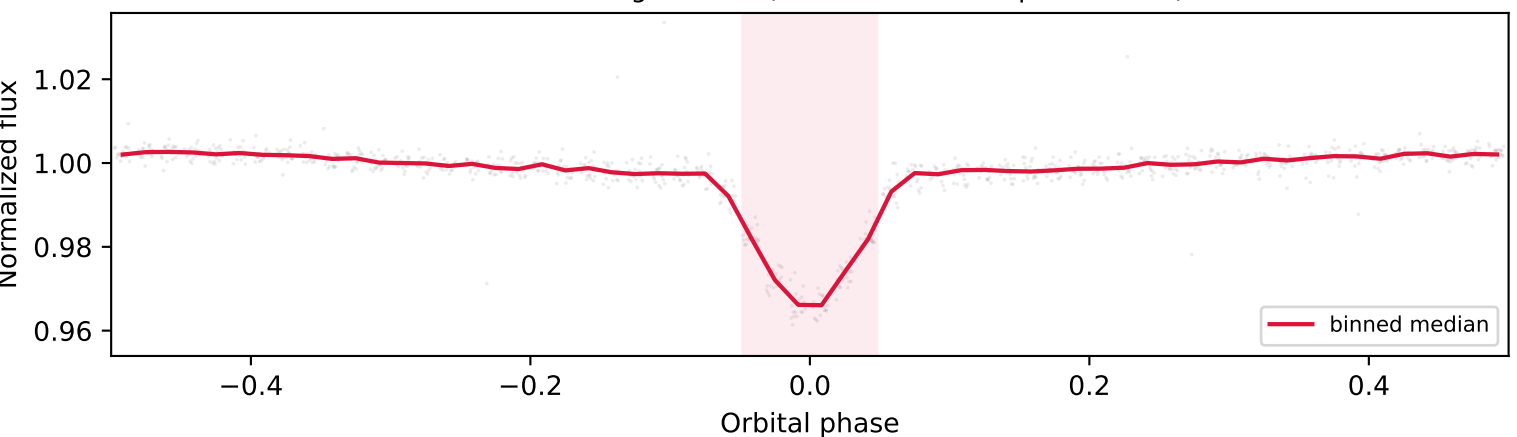
Eclipses observed 22 (gate ≥ 3)
Robust depth ratio 1.12 (gate ≥ 0.50)
Robust depth sigma 87.4 (gate ≥ 7.0)
Odd/even z-score 0.27 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 13 light-curve dip in this sector: 108.5 sigma t0 refined by -0.0050 phase
Difference-image peak SNR 15.52 (gate ≥ 5.0) Compactness 0.86 (gate ≥ 0.50)
Dimming centroid offset 0.14 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 365998896
2.59 px away, dTmag -1.63 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	1.906321 d	VSX	no match
Epoch (t0)	1625.2491 BTJD	ASAS-SN	no match
Duration	0.3500 d	Gaia eclipsing-binary table	no match
Depth	5.467 % (54,673 ppm)	Gaia variable classification	no match
SNR	202.5	TESS EB catalog (Prsa+2022)	no match
INDEPENDENT CHECKS (f3i gates)		ExoFOP TOI list	no match
Eclipses observed	14 (gate >= 3)	Verdict:	NOVEL
Robust depth ratio	1.08 (gate >= 0.50)		
Robust depth sigma	56.5 (gate >= 7.0)		
Odd/even z-score	0.02 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

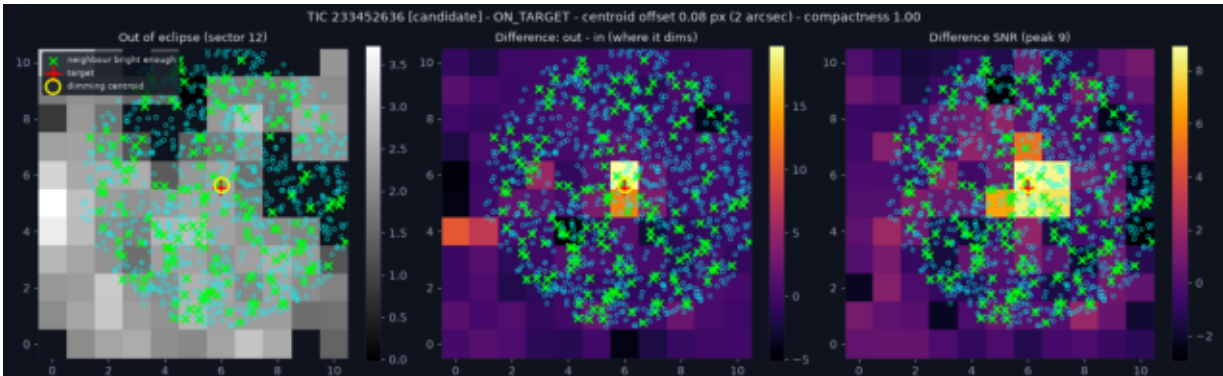
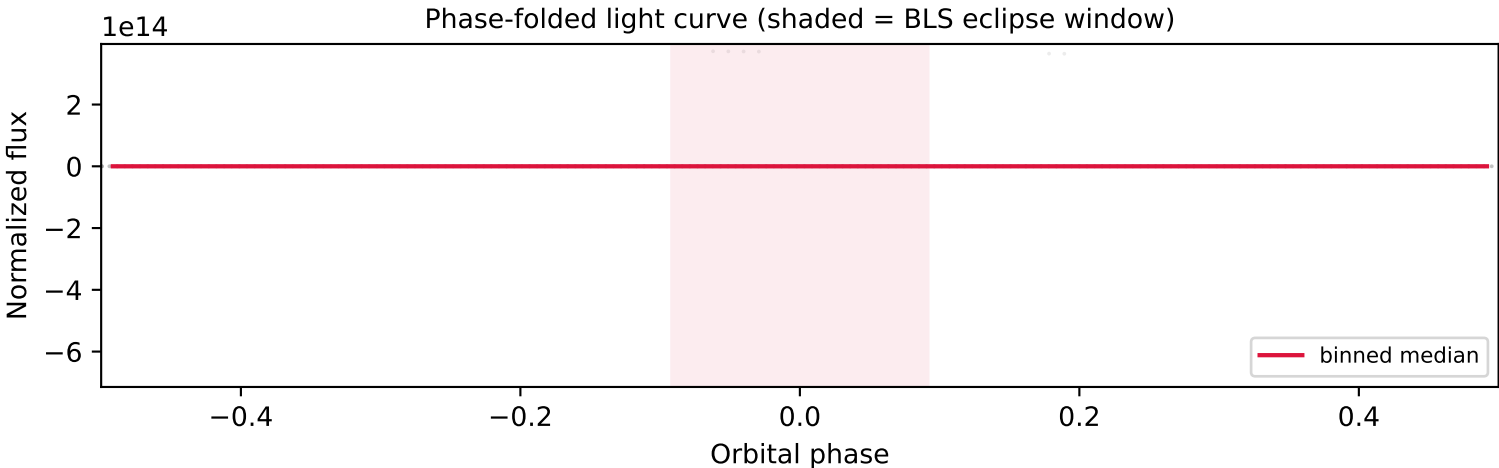
Sector 12 light-curve dip in this sector: 57.1 sigma t0 refined by 0.0050 phase

Difference-image peak SNR 8.94 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.08 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 1332737174

0.11 px away, dTmag +3.07 - farther from the dimming than the target



Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.030 px (dimming centroid 0.083 px from target vs 0.113 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.